

Week -7 PRATICAL

WEEK:7

Write a program that prints the addresses of code, global, static, heap, and stack variables. Analyze the Linux process address space layout and explain each memory segment.

Make use of the `/proc//maps` file and memory analysis tools to study the memory organization of a running process. Prepare a report explaining virtual memory mappings

```
#include <stdio.h>
#include <stdlib.h>
int global_var = 100;
static int static_var = 200;
void code_function()
{
    printf("Inside code_function()\n");
}
int main()
{
    int local_var = 10;
    int *heap_var = malloc(sizeof(int));
    if (heap_var == NULL)
    {
        printf("Memory allocation failed\n");
        return 1;
    }
    *heap_var = 300;
    printf("\n===== MEMORY ADDRESS DEMONSTRATION =====\n\n");
    printf("Code segment:\n");
    printf("Address of code_function : %p\n",
    (void *)code_function);
    printf("\nGlobal segment:\n");
```

```

printf("Address of global_var : %p\n",
(void *)&global_var);

printf("\nStatic segment:\n");

printf("Address of static_var : %p\n",
(void *)&static_var);


printf("\nHeap segment:\n");

printf("Address of heap_var : %p\n",
(void *)heap_var);

printf("\nStack segment:\n");

printf("Address of local_var : %p\n",
(void *)&local_var);

printf("\n===== \n");

free(heap_var);

return 0;

}

```

```

bharath@DESKTOP-6D34C2R:/mnt$ cd ~/PRATICAL/week7
bharath@DESKTOP-6D34C2R:~/PRATICAL/week7$ gcc -Wall -Wextra addresses.c -o addresses
bharath@DESKTOP-6D34C2R:~/PRATICAL/week7$ ls
addresses  addresses.c  memory_layout  memory_layout.c
bharath@DESKTOP-6D34C2R:~/PRATICAL/week7$ ./addresses

===== MEMORY ADDRESS DEMONSTRATION =====

Code segment:
Address of code_function : 0x60367b7ea1c9

Global segment:
Address of global_var : 0x60367b7ed010

Static segment:
Address of static_var : 0x60367b7ed014

Heap segment:
Address of heap_var : 0x603697982010

Stack segment:
Address of local_var : 0x7fffc33f7e2c

=====

```

Analyze Linux Process Address Space

```
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>

int global_var = 100;
static int static_var = 200;

int main()
{
    int stack_var = 300;

    int *heap_var = malloc(sizeof(int));

    if (heap_var == NULL)
    {
        perror("malloc");
        return 1;
    }

    *heap_var = 400;

    printf("Process ID: %d\n", getpid());

    printf("Address of code function: %p\n", (void *)main);
    printf("Address of global_var : %p\n", (void *)&global_var);
    printf("Address of static_var : %p\n", (void *)&static_var);
    printf("Address of heap_var : %p\n", (void *)heap_var);
    printf("Address of stack_var : %p\n", (void *)&stack_var);

    printf("\nProcess is running...\n");
    printf("Press ENTER to terminate.\n");

    getchar();

    free(heap_var);

    return 0;
}
```

```
bharath@DESKTOP-6D34C2R:~/PRATICAL/week7$ gcc -Wall -Wextra memory_layout.
-o memory_layout
bharath@DESKTOP-6D34C2R:~/PRATICAL/week7$ ./memory_layout
Process ID: 1222
Address of code function: 0x61d7e1bd7229
Address of global_var : 0x61d7e1bda010
Address of static_var : 0x61d7e1bda014
Address of heap_var : 0x61d7ee7a6010
Address of stack_var : 0x7ffdc67e6c

Process is running...
Press ENTER to terminate.
|
```

/proc/<PID>/maps

```
bharath@DESKTOP-6D34C2R:/$ cd ~/PRATICAL/week7
bharath@DESKTOP-6D34C2R:~/PRATICAL/week7$ cat /proc/1222/maps
61d7e1bd6000-61d7e1bd7000 r--p 00000000 08:30 36170 /home/bharath/PRATICAL/week7/memory_layout
61d7e1bd7000-61d7e1bd8000 r-xp 00001000 08:30 36170 /home/bharath/PRATICAL/week7/memory_layout
61d7e1bd8000-61d7e1bd9000 r--p 00002000 08:30 36170 /home/bharath/PRATICAL/week7/memory_layout
61d7e1bd9000-61d7e1bda000 r--p 00002000 08:30 36170 /home/bharath/PRATICAL/week7/memory_layout
61d7e1bda000-61d7e1bdb000 rw-p 00003000 08:30 36170 /home/bharath/PRATICAL/week7/memory_layout
61d7ee7a6000-61d7ee7c7000 rw-p 00000000 00:00 0 [heap]
70597d800000-70597d828000 r--p 00000000 08:30 13627 /usr/lib/x86_64-linux-gnu/libc.so.6
70597d828000-70597d9c0000 r-xp 00028000 08:30 13627 /usr/lib/x86_64-linux-gnu/libc.so.6
70597d9c0000-70597da0e000 r--p 001c0000 08:30 13627 /usr/lib/x86_64-linux-gnu/libc.so.6
70597da0e000-70597dal2000 r--p 0020d000 08:30 13627 /usr/lib/x86_64-linux-gnu/libc.so.6
70597dal2000-70597da14000 rw-p 00211000 08:30 13627 /usr/lib/x86_64-linux-gnu/libc.so.6
70597da14000-70597da21000 rw-p 00000000 00:00 0
70597da77000-70597da7a000 rw-p 00000000 00:00 0
70597da7f000-70597da81000 rw-p 00000000 00:00 0
70597da81000-70597da82000 r--p 00000000 08:30 13624 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
70597da82000-70597dab1000 r-xp 00001000 08:30 13624 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
70597dab1000-70597dabc000 r--p 00030000 08:30 13624 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
70597dabc000-70597dabe000 r--p 0003b000 08:30 13624 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
70597dabe000-70597dabf000 rw-p 0003d000 08:30 13624 /usr/lib/x86_64-linux-gnu/ld-linux-x86-64.so.2
70597dabf000-70597dac0000 rw-p 00000000 00:00 0
7ffdcaf48000-7ffdcaf6a000 rw-p 00000000 00:00 0 [stack]
7ffdcafe4000-7ffdcafe8000 r--p 00000000 00:00 0 [vvar]
7ffdcafe8000-7ffdcafea000 r-xp 00000000 00:00 0 [vdso]
bharath@DESKTOP-6D34C2R:~/PRATICAL/week7$ |
```

```
7ffdcafe8000-7ffdcafea000 r-xp 00000000 00:00 0 [vdso]
bharath@DESKTOP-6D34C2R:~/PRATICAL/week7$ cat /proc/1222/maps | grep heap
61d7ee7a6000-61d7ee7c7000 rw-p 00000000 00:00 0 [heap]
bharath@DESKTOP-6D34C2R:~/PRATICAL/week7$ cat /proc/1222/maps | grep stack
7ffdcaf48000-7ffdcaf6a000 rw-p 00000000 00:00 0 [stack]
bharath@DESKTOP-6D34C2R:~/PRATICAL/week7$ cat /proc/1222/maps | grep memory_layout
61d7e1bd6000-61d7e1bd7000 r--p 00000000 08:30 36170 /home/bharath/PRATICAL/week7/memory_layout
61d7e1bd7000-61d7e1bd8000 r-xp 00001000 08:30 36170 /home/bharath/PRATICAL/week7/memory_layout
61d7e1bd8000-61d7e1bd9000 r--p 00002000 08:30 36170 /home/bharath/PRATICAL/week7/memory_layout
61d7e1bd9000-61d7e1bda000 r--p 00002000 08:30 36170 /home/bharath/PRATICAL/week7/memory_layout
61d7e1bda000-61d7e1bdb000 rw-p 00003000 08:30 36170 /home/bharath/PRATICAL/week7/memory_layout
bharath@DESKTOP-6D34C2R:~/PRATICAL/week7$ grep -E '\[heap\]|\[stack\]' /proc/1222/maps
61d7ee7a6000-61d7ee7c7000 rw-p 00000000 00:00 0 [heap]
7ffdcaf48000-7ffdcaf6a000 rw-p 00000000 00:00 0 [stack]
bharath@DESKTOP-6D34C2R:~/PRATICAL/week7$ |
```